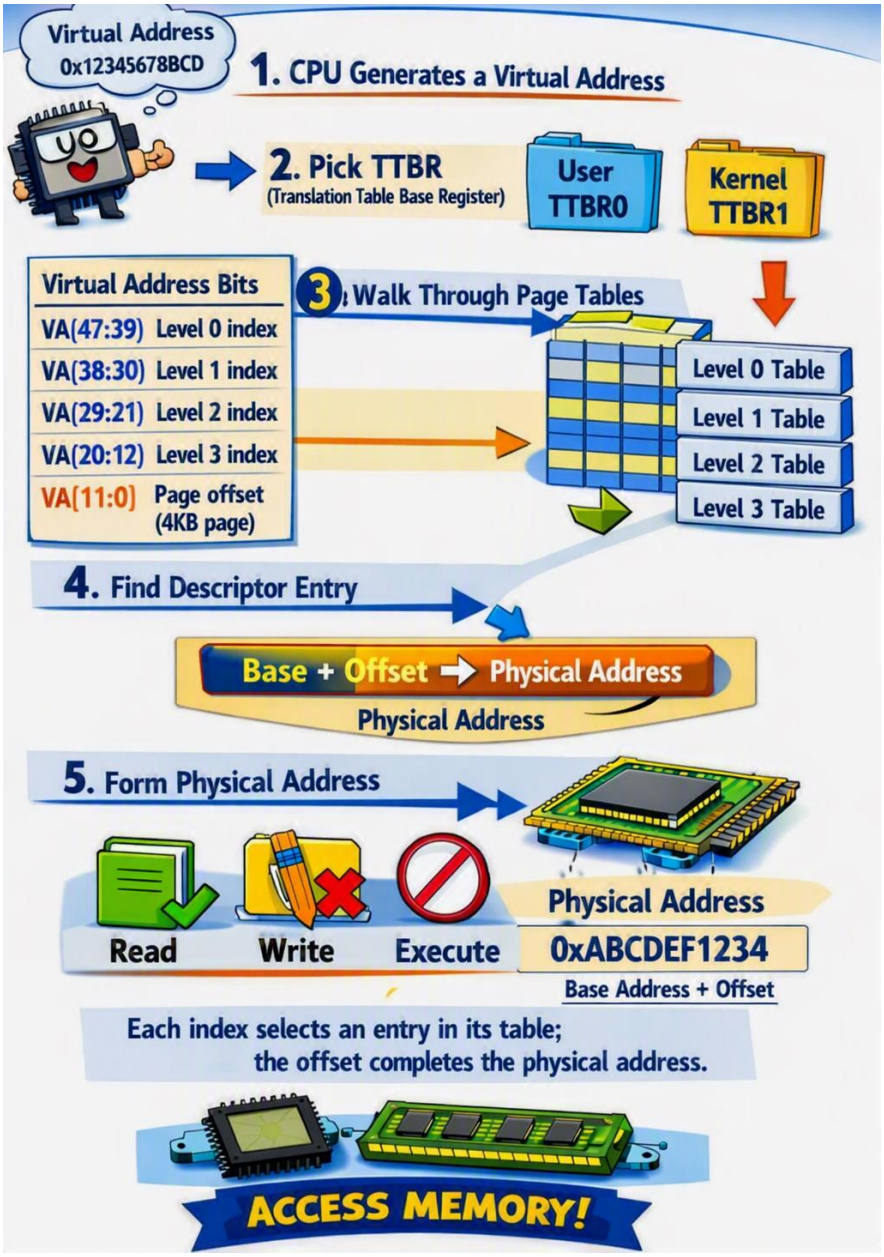
MMU Descriptor Bit-Level Summary

(ARMv8-A, 4KB Granule)



1. Block Descriptor (L1 / L2) — Bit-Wise Description

`Descriptor Bits (Block: 1GB at L1, 2MB at L2)`

[63:55] Reserved / implementation-defined
[54] UXN (Unprivileged Execute Never)
[53] PXN (Privileged Execute Never)
[52] Contiguous hint
[51] DBM (Dirty Bit Modifier)
[50:48] Reserved
[47:12] Output address (physical base of block)
[11] nG (not global)
[10] AF (Access Flag)
[9:8] SH (Shareability: 00 NS, 10 Outer, 11 Inner)
[7:6] AP (Access permissions)
[5] NS (Non-secure)
[4:2] AttrIndx (MAIR attribute index)
[1:0] Descriptor type = 01 (Block)

2. Page Descriptor (L3) — Bit-Wise Description

`Descriptor Bits (Page: 4KB)`

[63:55] Reserved / implementation-defined
[54] UXN (Unprivileged Execute Never)
[53] PXN (Privileged Execute Never)
[52] Contiguous hint

[51] DBM (Dirty Bit Modifier)
[50:48] Reserved
[47:12] Output address (physical base of 4KB page)
[11] nG (not global)
[10] AF (Access Flag)
[9:8] SH (Shareability)
[7:6] AP (Access permissions)
[5] NS (Non-secure)
[4:2] AttrIndx (MAIR attribute index)
[1:0] Descriptor type = 11 (Page)

3. Table Descriptor (L0 / L1 / L2) — Bit-Wise Description

```
Descriptor Bits (Table: points to next-level table)
```

[63:52] Reserved / implementation-defined
[51] DBM (Dirty Bit Modifier)
[50] APTable
[49] UXNTable
[48] PXNTable
[47:12] Next-level table base address (physical address)
[11:2] Reserved
[1:0] Descriptor type = 11 (Table)

```
Notes:
- No AttrIndx
```

4. Virtual Address Bit-Wise Description (48-bit VA, 4KB Granule)

VA[47:39]	Level 0 index (9 bits)
VA[38:30]	Level 1 index (9 bits)
VA[29:21]	Level 2 index (9 bits)
VA[20:12]	Level 3 index (9 bits)
VA[11:0]	Page offset (4KB page)

Properties:

- Each index = 9 bits → 512 entries per table
- VA contains only indices + offset
- Descriptor bits are NOT part of the VA